



Business Case, Part 1: Background and Staging

Agile Data Engine for Data Engineers — AI-
Assisted Development





PackageDelivery

PackageDelivery is a delivery company in New York City.

It was planning to expand into taxi operations.

Before deciding, it started a market analysis with open NYC taxi data.





The taxi data

Source: NYC Taxi Data



Yellow taxi trips



Green taxi trips



Taxi zones

Zone IDs, zones and boroughs



Rate codes

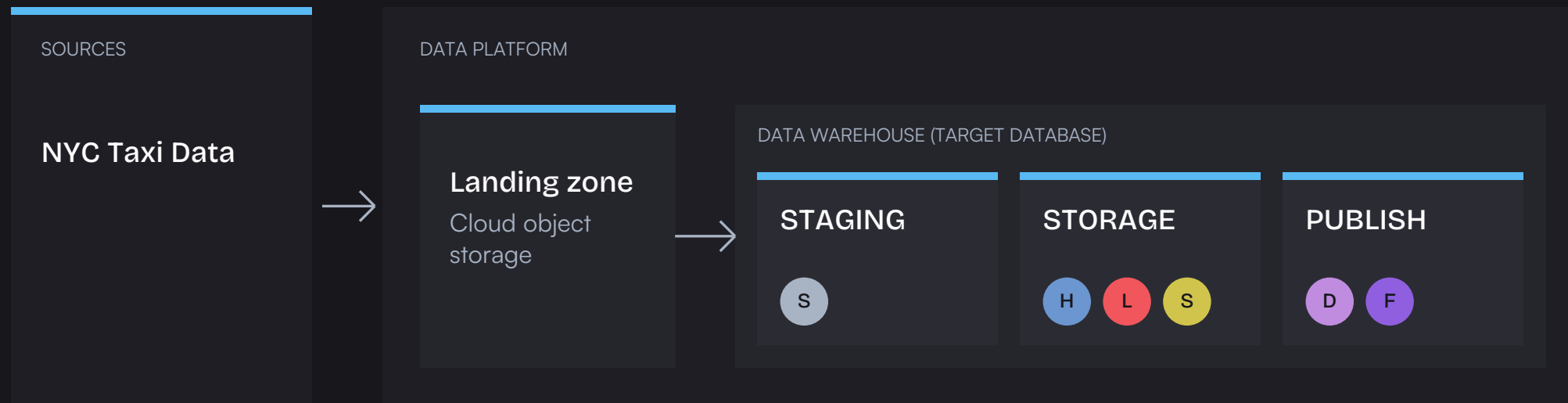


What was built in basic training

Staging: the files loaded as they are, to test whether the data is useful.

Storage: a Data Vault with hubs for taxi zones, trips and rate codes, plus links and satellites.

Publish: a dimensional model for reporting. Trips are a fact; zones and rate codes are dimensions.





How it was built

01

By hand

In Agile Data Engine's Designer

02

One layer at a time

Then a new dataset from source to publish

03

Small, tested steps

Incremental development, each step deployed to production



The decision was left open

The data warehouse can show where taxis drive and what they earn.

But should PackageDelivery start taxi operations — and where?

That question was never answered.



What the taxi data cannot tell

The market

Taxi data shows the market



Our capacity

It does not show whether PackageDelivery has anything to offer it

For that, PackageDelivery needs to know its own fleet





The idea

Could idle vans carry passengers, where taxi prices are good and the fleet is already present?

1.

Morning

Delivery vans are busy

2.

Midday

Some vans stand idle

3.

Afternoon

Delivery vans are busy



The new source: FLEETOPS

FLEETOPS is PackageDelivery's fleet management system. It exports three files:

FILE

ONE ROW PER

fleet_vehicle.csv

Vehicle: type, passenger seats, cost per mile, home borough

fleet_shift.csv

Driver shift: vehicle, depot, start and end time

fleet_movement.csv

Movement of a vehicle from one taxi zone to another



About the fleet data

January—February 2017, the same period as the taxi data.

42 vehicles in New York's five boroughs.

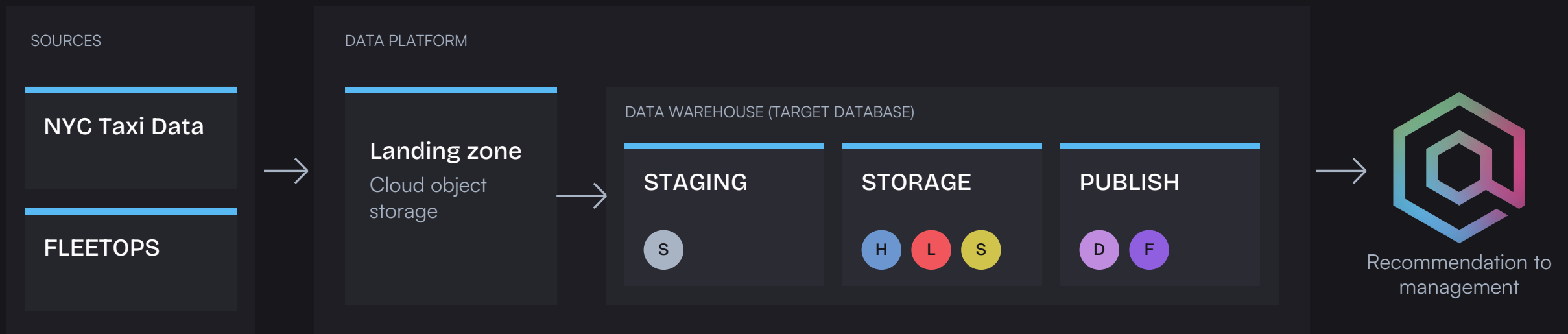
Different vehicle types. Not all of them can carry passengers.

Locations use the same taxi zone IDs as the taxi data.



What happens now

The same data warehouse, extended with FLEETOPS.
This time ADA builds. You specify, review and verify.
At the end: a recommendation to PackageDelivery's management.





Exercises 1 and 2

Exercise 1: install ADA, connect it to Agile Data Engine, and bring FLEETOPS into staging.

Exercise 2: when a load fails, find the cause, decide how to handle it, and fix it.

